NEAR AND FAR-FIELD DEFORMATION FROM THE 2023 TURKEY EARTHQUAKES USING GNSS DATA.

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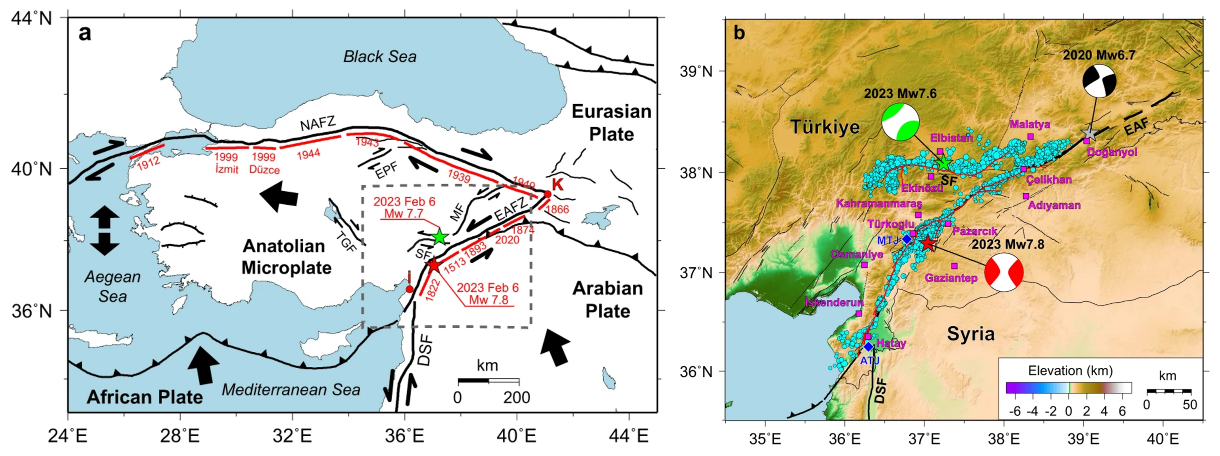
INTRODUCTION

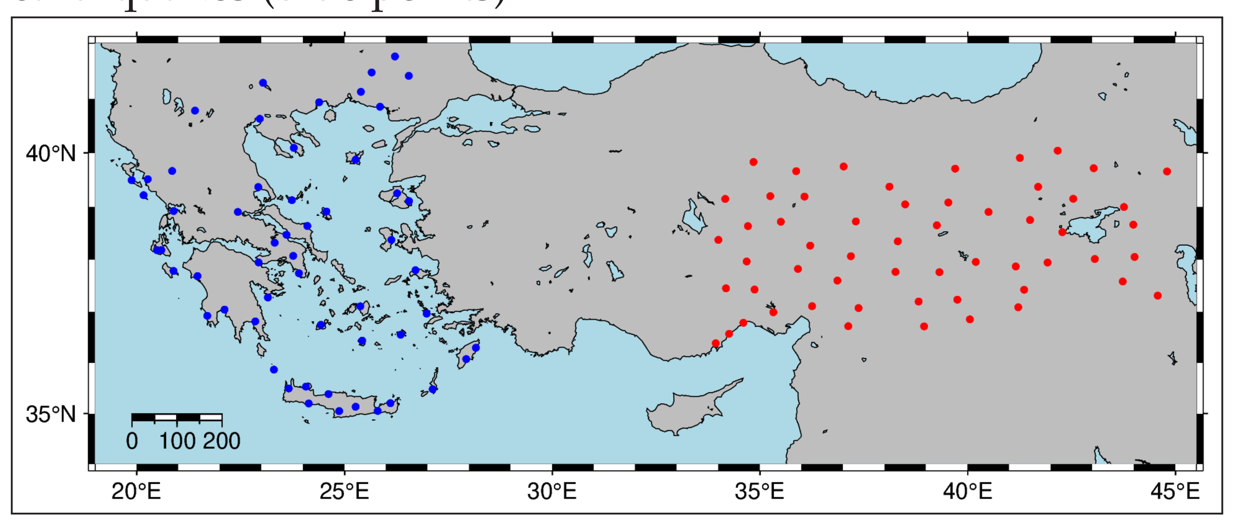
On February 6th, 2023 at 01:17:35 UTC the M_w 7.8 Nurdağı-Pazarcık earthquake nucleated ∼15 km southeast of the mapped trace of the East Anatolian Fault Zone. Relocations place the hypocenter at (37.0234° E, 37.2444° N, depth=12 km) and analyses of teleseismic data show a left-lateral source mechanism on a vertical or near vertical fault. A vigorous aftershock sequence followed a little over 9 hours after the first event, at 10:24:49 UTC, the M_w 7.6 [1]



Tectonic backround (a) and focal mechanisms (b) of 2023 Turkey earthquake doublet [2].

GNSS STATIONS

A network of 95 continuous GNSS stations was utilized for this study, comprising: a) 41 continuously operating stations located near the earthquake sequence in southwest Turkey (<700km distance from epicentre, red points) and b) 54 far-field stations distributed across the Aegean Sea and mainland Greece, located 800 to 1600 km from the epicentres of the two earthquakes (blue points).



Acknowledgments *to Metrica SA for the free availability of the 1Hz GNSS data of HxGN SmartNet for this study, and TUSAGA-Aktif provide GNSS data of Turkish GNSS stations*

GNSS DATA PROCESSING

Data collected from the selected network of 95 GNSS sites were processed using the well known PPP-AR approach, in pseudo-kinematic mode, estimating site cordinates at every measurement epoch. Below, we summarize analysis options and products used:

Processing scheme:

- Precise Point Positioning with Ambiguity Resolution using PRIDE-PPPAR Software (v3.1.2) [3].
- Satellite systems: GPS / GLO / GAL / BDS-2/3
- Elevation angle: 7 deg
- Sampling interval: 1s
- Coordinates estimated in pseudo-kinematic mode
- Reference Frame: IGS20
- Final IGS Products for satellite orbits and EOPs
- ANTEX File: IGS20_2247
- Tropospheric modeling: VMF1

TIME SERIES ANALYSIS

1. Pre-processing and Filtering of GNSS Data

Co-seismic displacements were estimated only for GNSS stations located within 800 km of the earthquake epicenters. For stations situated at distances greater than 800 km, no significant co-seismic displacements were expected.

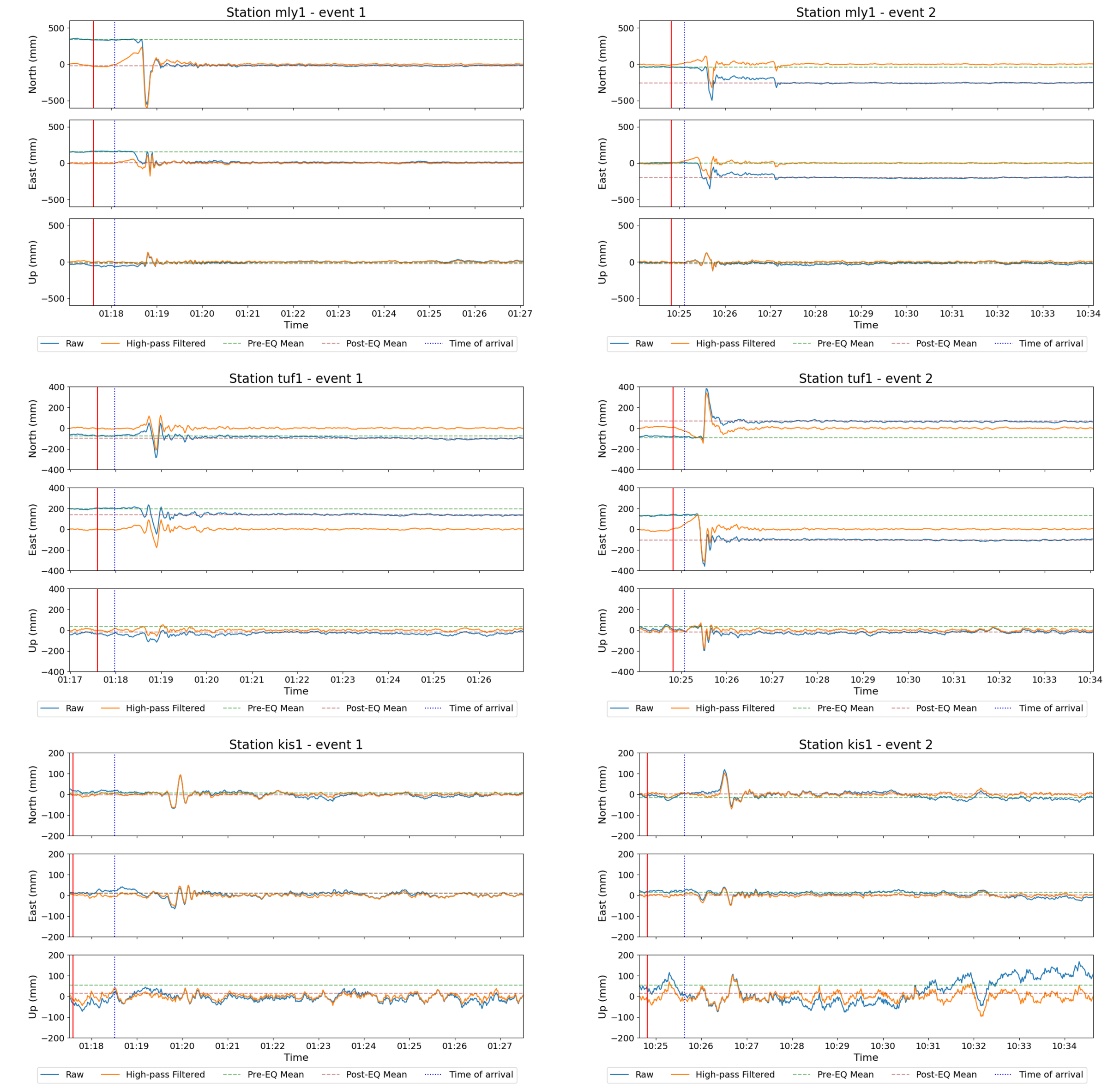
Low-frequency noise in the GNSS data—likely caused by multipath effects, unmitigated tropospheric delays, and other sources—was suppressed using a high-pass Butterworth filter with a cut-off frequency of 0.1 Hz. This filtering effectively eliminated slow-varying noise components without significantly altering potential dynamic displacements

2. Peak Ground Displacement and Noise Estimation

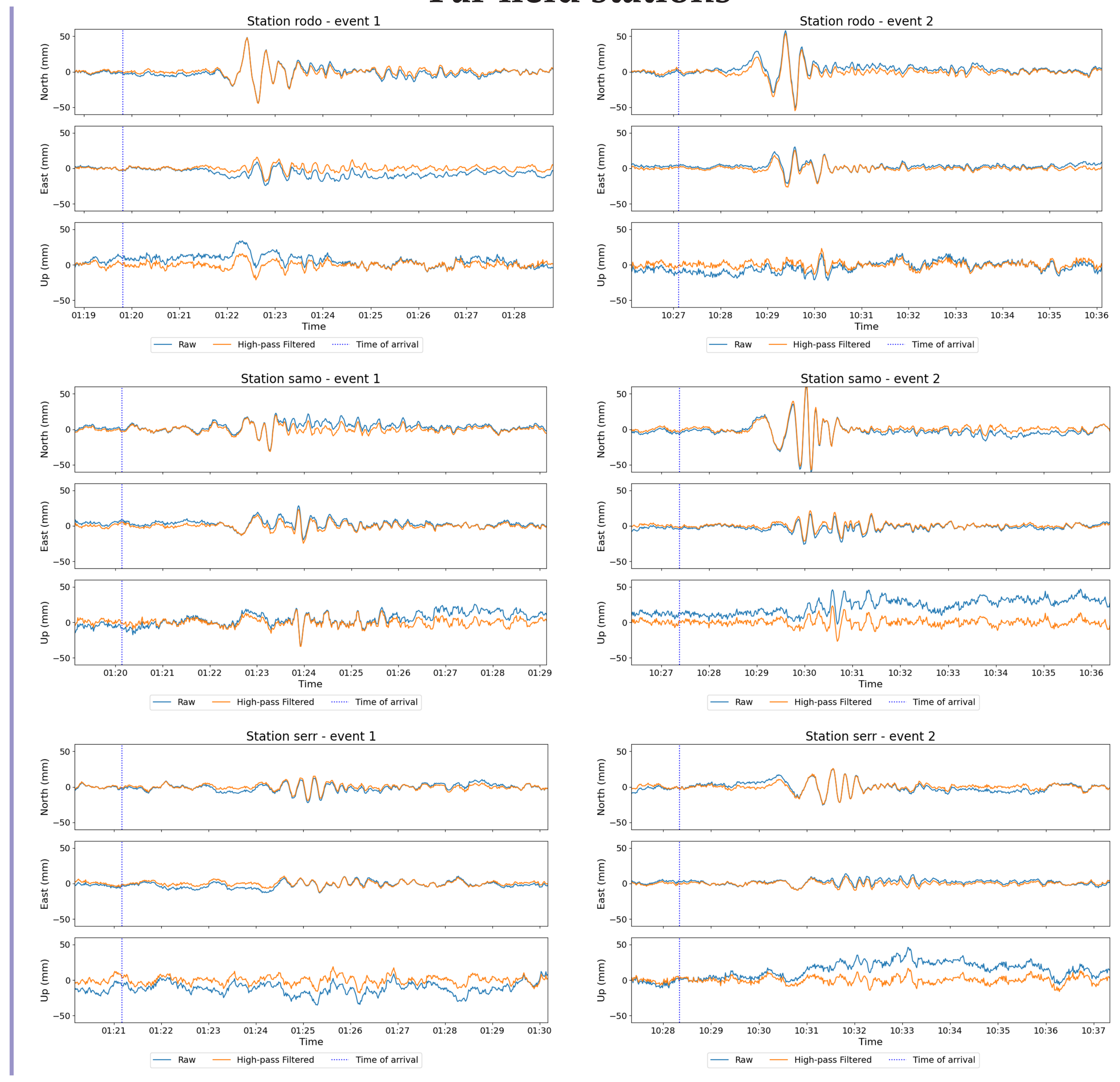
The transient seismic response window for each GNSS station was defined as starting at the arrival time of the P-waves—calculated assuming a velocity of 6 km/s—and spanning 400 seconds to ensure the full signal was captured. Peak Ground Displacement (PGD) in the NEU components was identified as the maximum value within this interval. To estimate the noise level, we calculated the standard deviation of the filtered neu time series over a 10-minute window prior to the first seismic event. A conservative threshold of 3σ was used to distinguish real transient responses from background noise.

ANALYSIS RESULTS

Near-field stations

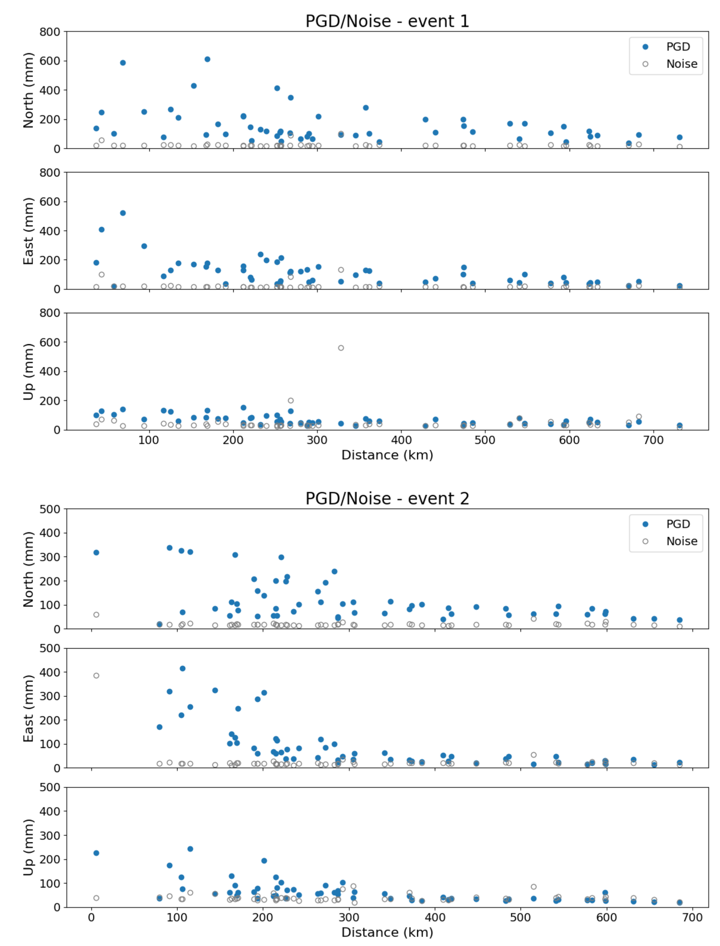


Far-field stations

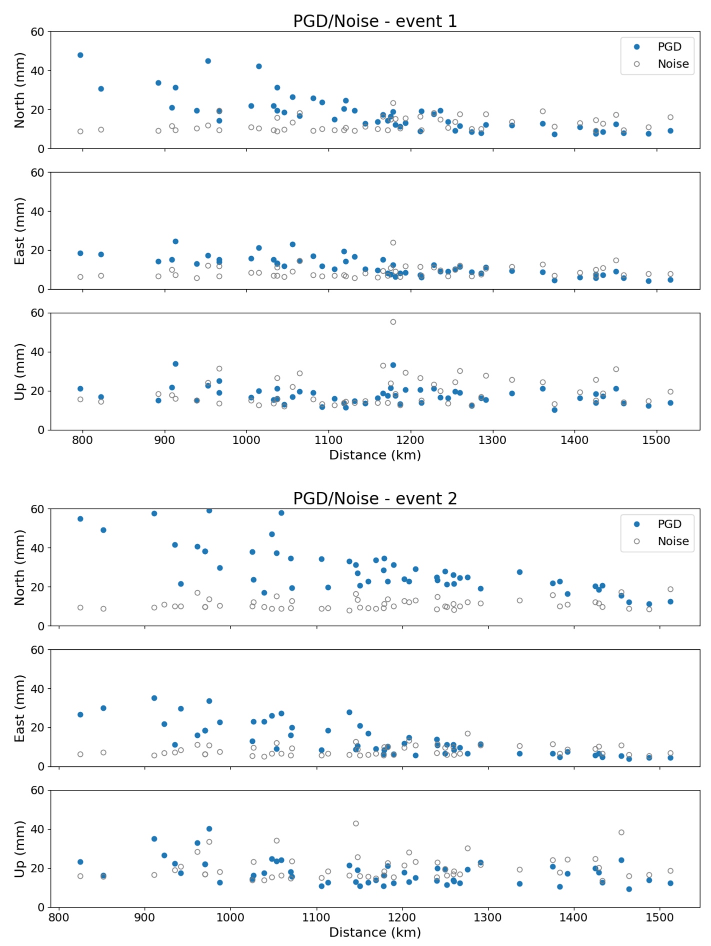


DISCUSSION

Near-field



Far-field



Preliminary results from PGD values:

- The highest PGD values (up to 60 cm) are observed at near-field stations located within 300 km of the epicenters.
- PGD values generally decrease with distance from the epicenters. However, due to the directivity of seismic wave propagation and fault orientation.
- The highest PGD values are observed at the North component for both seismic events, indicating that the North-South direction is mostly affected by the propagation of the seismic waves.
- The second seismic event has a larger far-field impact, probably due to the geometry of the seismic fault and the directivity of the seismic waves propagation.
- The Up component is not affected by the seismic events potentially due to the small impact that the S-waves may have for far-field stations.

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Photography encouraged

FUTURE RESEARCH

- Investigate the co-seismic displacements of the near-field GNSS stations.
- Investigate the far field of the directivity effect of the propagation seismic waves through the PGD of the GNSS stations.
- Investigate the frequency context of the seismic waves and its potential variation with the distance from the epicentre.
- Estimate of the seismic waves velocity as detected through the network GNSS stations displacement.

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